

RABIJITH K R

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linkedin.com/in/rabijith-k-r — github.com/Rabijith-KR — Portfolio Website

Professional Summary

Electronics and Communication Engineering graduate with experience in IoT, embedded systems, and AI-assisted web development. Skilled in Python, SQL, Arduino, ESP32, Git, and GitHub, with hands-on experience using AI-assisted and agentic coding tools (Antigravity) to design, build, and deploy full web applications. Strong problem-solving abilities with proven project execution.

Technical Skills

- **Languages:** Python, SQL
- **Embedded & IoT:** Arduino, ESP32, RFID, BMP280, DS18B20, Peltier Module
- **Tools:** Git, GitHub, VS Code, Arduino IDE, EmailJS
- **AI-Assisted Development:** Antigravity, Agentic Coding Workflows, AI-Directed Web Development
- **Concepts:** Embedded Systems, IoT Systems, Responsive Web Design, Requirement-Driven Development

Projects

IoT-Based Voice-Announcing UPI Payment Terminal *Technologies: ESP32, C++, Razorpay API, MQTT, Node.js, Render, TFT Display, I2S Audio*

- Built an ESP32 payment terminal that receives real-time UPI webhooks and announces any transaction amount via dynamically synthesized speech.
- Integrated Razorpay (webhook verification, dynamic QR generation), a Node.js backend on Render, and MQTT for real-time device communication.
- Diagnosed and resolved critical hardware/firmware faults — including a defective touch controller and a driver-level ADC conflict — through systematic testing rather than trial-and-error.

Responsive Interactive Website *Technologies: Antigravity (AI-Assisted Development), GitHub Pages Live Demo*

- Designed and built a responsive interactive website featuring animations, music integration, galleries, quizzes, timelines, and countdown timers using AI-assisted, agentic coding tools (Antigravity).
- Directed AI-driven implementation of dynamic UI interactions, including typing effects, image lightboxes, confetti animations, and interactive message cards, through iterative prompting and testing.
- Reviewed, debugged, and refined AI-generated code to ensure functionality and responsiveness across devices.

Authenticated Voting System using RFID and Fingerprint *Technologies: C++, Arduino, RFID Module, Fingerprint Sensor*

- Developed a secure electronic voting prototype using RFID and fingerprint authentication.
- Implemented dual-layer voter verification and duplicate-vote prevention logic.
- Performed hardware interfacing, testing, and system validation.

Solar-Powered Portable Smart Medicine Storage & Dispensing System *Technologies: ESP32, RFID, DS18B20, LCD Display, Peltier Module*

- Developed a portable smart medicine storage system with dual-temperature compartments.
- Integrated RFID authentication and real-time temperature monitoring using ESP32 and DS18B20.
- Designed an LCD interface displaying medicine details and system parameters.

Internship Experience

Engineering Intern 30 July 2026 - 30 AUG 2026 *Serendipity Labs (OPC) Pvt. Ltd.*

- Project: IoT-Based Voice-Announcing UPI Payment Terminal

Industry Study Intern Jan 2026 – Feb 2026 *TCS BFSI Digital Garage, Trivandrum*

- Contributed to the development of an IoT-based assistive technology prototype.
- Participated in solution design, testing, and industry-oriented project workflows.

Education

B.Tech in Electronics and Communication Engineering 2022 – 2026 College of Engineering Chengannur
CGPA: 7.55

Certifications

- Python Mastery – Udemy